PNAS RESEARCH ARTICLE | CLASSIFICATION

Forecasting S&P 500 Trends using Graph Attention Networks on Causal Structures

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This manuscript was compiled on March 12, 2026

Predicting stock prices is challenging due to the complex interactions between market participants, temporal trends, and sudden market shocks/world events. In this study, we compare several models for one-step-ahead stock price prediction, including a Linear Regression (LR) model, Long Short-Term Memory (LSTM), Graph Convolutional Network (GCN), combined LSTM and GCN, Graph Attention Network, and Attention Graph Neural Network (GNN). All models were trained on normalized historical stock prices, with graph-based models leveraging pre-constructed stock relationship networks derived from Pearson correlation, Lift, and Granger causality. Our results show that the Attention GNN achieved the lowest mean squared error (0.0087), outperforming other models by allowing edge-specific attention weights that adaptively capture the influence of related stocks. The LSTM-GNN (0.0110) also performed well, demonstrating the value of combining temporal dynamics with network-based relationships. Standard GAT (0.0116) benefited from the network structure but was constrained by k-limited edges, while the lone LSTM (0.0132) outperformed the GNN (0.0137) by capturing individual stock trends. The LR model (0.1071) performed poorly, highlighting the inadequacy of simple linear approaches. These findings underscore the importance of modeling both temporal patterns and cross-stock interactions for accurate stock price prediction.

Stocks | Technology | Linear Regression | GNN | LSTM | GAT

Significance

Accurately predicting stock prices is critical for investors, portfolio managers, and financial institutions to make informed decisions and manage risk. This study demonstrates that the combination of temporal modeling with inter-stock relationships via combining Long Short-Term Memory and Graph Convolutional Networks, along with attention-based Graph Neural Networks, significantly improves predictive accuracy over traditional models. By explicitly capturing both historical trends and the varying influence of related stocks, these models provide a more nuanced understanding of market dynamics, offering practical insights for financial forecasting and algorithmic trading strategies.

Introduction

Accurate stock price prediction remains a fundamental challenge in finance due to the complex, dynamic, and interconnected nature of financial markets. Traditional models, such as linear regression, often fail to capture nonlinear dependencies and temporal trends inherent in stock price movements. Recent advances in machine learning have introduced models that leverage both temporal patterns and relationships between stocks, offering improved predictive capabilities.

Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks (1), are well-suited for modeling sequential data and capturing historical trends for individual stocks. However, these models typically do not account for inter-stock relationships that can influence a stock's future behavior. Graph-based models, such as Graph Convolutional Networks (GCNs) (2) and Graph Attention Networks (GATs) (3), incorporate stock relationships into prediction, allowing each stock to aggregate information from related stocks in a structured network. Attention mechanisms further enhance these models by enabling edge-specific weighting, which allows the model to learn the varying influence of each connected stock.

In this study, we compare several approaches for one-step-ahead stock price prediction, including linear regression, LSTM, GNN, LSTM-GNN, GAT, and Attention GNN. Using historical stock prices and networks constructed from correlation and Granger causality, we evaluate the predictive performance of each model to understand the benefits of capturing temporal dynamics and cross-stock dependencies.

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F.G., L.L., C.S., and D.V. designed research; F.G. and L.L performed research; H.W, L.L., and C.S. performed analysis; and D.V. wrote the paper.

To facilitate further research, we publicly release our code and dataset at: https://github.com/Lieqi-Liu/Math-168

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Dataset

The dataset was obtained from Yahoo Finance through an API endpoint used to retrieve historical market data. (4) The dataset contains adjusted closing prices for 13 technology stocks spanning 3,270 trading days between January 2, 2013, and December 31, 2025. The stocks included are AAPL, AMZN, GOOGL, JNJ, JPM, META, MSFT, NVDA, PG, XOM, TSM, AVGO, and ORCL.

Data Preprocessing. The dataset was free of NA and missing values, so data cleaning was not needed. The preprocessing began by converting the dates column into an ordinal scale, ranging from 1 to 3270. The next step was to split the dataset into training and test periods, with the last 30 days of data reserved for testing. Each stock's values were min–max scaled based on the training set using $x_{scaled} = \frac{x-min(x)}{max(x)-min(x)}$. Stock prices were also converted into daily returns (percentage changes) to stabilize variance in different stock prices using $\frac{P_t-P_{t-1}}{P_{t-1}}$.

Network Creation. Using the daily returns, a Pearson correlation matrix was created, and high correlation(> 0.5) relationships were linked with an edge. The data was then discretized into binary values (1 for positive return, 0 otherwise). These binary values were then used to calculate Lift, which measures how much more stocks move up together given they are independent using the formula $Lift(A, B) = \frac{P(A \wedge B)}{P(A)*P(B)}$. Any stocks with high Lift (> 1.3) had an edge drawn between them. These 2 edge generation methods were used to create a stock relationship network.

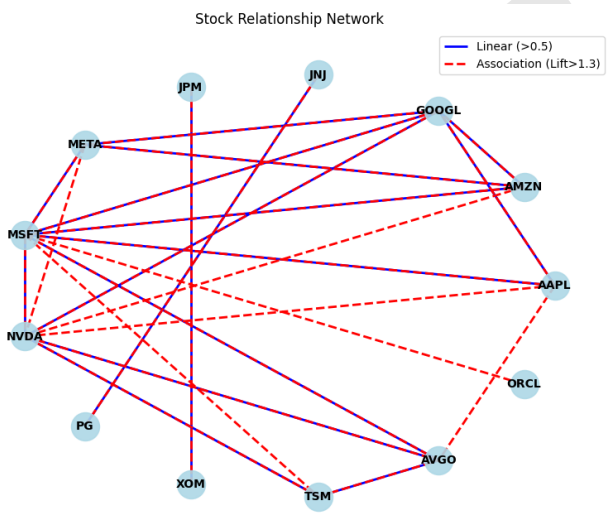

Fig. 1. Stock relationship network generated from correlation and lift metrics.

Methods

Each model was trained to minimize the Mean Squared Error(MSE) between its predictions and the actual values on the training dataset, and its performance was evaluated using the mean squared error on a separate test dataset.

Linear Regression (LR) Model. This model was used as a baseline for comparison with more complex models. An LR model was trained separately for each stock using the raw training data. The trained models were then used to predict prices in the testing dataset. Because each model was fit independently, it captured only the temporal behavior of individual stocks.

Long Short-Term Memory (LSTM) Model. This model only used historical stock prices to make predictions. This model used the data normalized using min–max scaling. The model functions by iterating through the dataset 1 day at a time and predicting the scaled stock return based on the last 5-day window. As it iterates, it learns temporal patterns such as trends or momentum. The model learned from the training dataset in batch sizes of 256 for 100 epochs before making a prediction for each individual stock in the testing dataset.

Graph Convolutional Network (GCN) Model. This model used the data normalized with min–max scaling along with the stock relationship network to predict stock price. The network was converted into a normalized adjacency matrix and used within a two-layer GCN network. At each time step (1 day), the model aggregates information from neighboring stocks (connected edges), including the stock itself, and transforms this information using learned weight matrices. The resulting representation is used to predict the price of each stock on the following day. The model was trained for 100 epochs using batches of size 256.

LSTM and GCN Model. This model utilized the normalized min–max scaled data along with the stock relationship network to predict stock price. This model combined the LSTM and GNN models. The model iterates through the data 1 day at a time, like before. First, the model uses LSTM to look at each individual stock's 5-day past and turns each of the pasts into a feature vector. Then, rather than the GNN using the prices of the current day to create a weighted matrix, it uses the 5-day feature vectors, allowing the model to absorb more information at each iteration. This combines LSTM's ability to learn patterns of stock movements with GCN's ability to look at the interactions between different stocks. The model was trained for 100 epochs using batches of size 256.

Attention Graph Neural Network (GNN) Model. This model used the data normalized using min–max scaling along with the stock relationship network to predict stock prices. The model is a variation on the GCN model. In the GCN model, the contribution of neighboring stocks is determined by the normalized adjacency matrix, which assigns fixed weights based on the graph structure. The Attention GNN extends this approach by learning attention weights for each connected edge. These attention weights allow the model to assign different levels of importance to neighboring stocks when predicting a given stock's price. As a result, some connected stocks may have a stronger influence on the prediction than others. The model was trained for 100 epochs using batches of size 256.

Granger Causality and Graph Attention Network (GAT) Model. This model uses data normalized via min–max scaling along with a directed stock relationship network derived from Granger causality tests on historical returns. The model

examines which neighbors' pasts have the largest influence on a stock's current price, and if it is strong enough ($p < .05$), a node is created. For each stock, only the top-k strongest Granger-causal stocks are considered as neighbors, forming a directed adjacency matrix. The model employs a two-layer GAT where each stock aggregates information from its neighbors with learned attention weights, similarly to the Attention GNN, allowing the model to dynamically weigh the influence of each connected stock when predicting the next-day price. The model was trained for 100 epochs using batches of 256 samples.

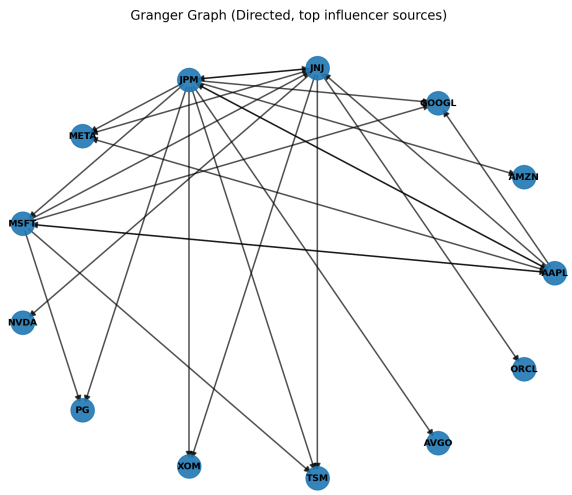

Fig. 2. Stock relationship network generated from Granger correlation.

Results

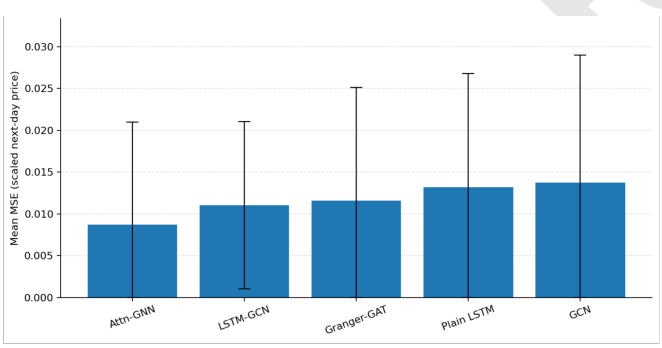

Fig. 3. Stock relationship network generated from correlation and lift metrics.

Attention GNN achieved the lowest mean MSE (0.0087) which demonstrates that allowing edge-specific attention weights improves predictive accuracy. The LSTM-GNN (0.0110) also performed very well, indicating that capturing both individual stock trends over time and the influence of related stocks is important for accurately predicting future stock prices. GAT (0.0116) performed well but worse than Attention GNN showing that looking at other stocks in very valuable in prediction but perhaps the Granger Causality network is lacking. Lone LSTM (0.0132) did slightly worse than the other models but did better than GNN (0.0137) indicating that learning trends in stock movements can be better than predicting based on neighbors if the weights of each edge are not properly calibrated. The Linear model had the highest mean MSE (0.1071), confirming that simple linear relationships cannot capture the complex dynamics in stock prices.

Conclusions and Discussion

In this study, we compared multiple neural network-based models for predicting daily stock prices, including a standard LSTM, a GCN, an Attention-based GNN, a GAT, and a hybrid LSTM-GNN.

The results indicate that models incorporating cross-stock relationships consistently outperform models that consider each stock in isolation. In particular, the Attention GNN achieved the lowest mean squared error (0.0087), demonstrating the importance of modulated relational information. The LSTM-GCN combination also performed strongly (0.0110), highlighting the value of attention mechanisms in inter-stock dependencies and learning broader market trends. Granger-Gat used a similar methodology to Attention GNN yet performed worse (0.0116) showcasing the importance of network creation. Standard GCN and LSTM models performed comparatively worse, illustrating that static aggregation of neighbors or reliance on past market trend is not enough to correctly predict future movements. Finally, the LR Models abysmal performance (0.1071) confirms that linear relationship are ill equipped to predict the stock market.

Overall, these findings suggest that effective stock prediction benefits from models that simultaneously account for both temporal patterns and networked relationships among stocks. Future work could explore multi-head attention, larger graph structures, or the integration of alternative market signals to further improve predictive accuracy.

Limitations

While the results demonstrate that graph-based neural networks can improve stock price prediction, several limitations should be noted. First, the models rely on historical price data and pre-defined relationships between stocks (e.g., Pearson correlation, Granger causality), which may not capture sudden market shocks or events that break historical patterns. Second, the choice of hyperparameters—such as the number of neighbors in the GNN, LSTM sequence length, or attention dropout—can significantly impact performance as the amount of stocks in a network grows. Third, the study focuses on technology stocks with results perhaps not being generalizable to all sectors. Finally, while the Attention GNN achieved the lowest mean MSE, this improvement is modest in some cases, and predictive performance may still be insufficient for real-world trading without additional features such as macroeconomic indicators or news sentiment. Future work could explore integrating these data sources and evaluating robustness across multiple market conditions and sectors.

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